

1] What are constraints in SQL? List and explain the different types of constraints.

Ans:

What are Constraints in SQL:

Constraints in SQL are rules applied to columns in a table to enforce data integrity, accuracy, and reliability. They ensure that the data entered into a table meets specific criteria and prevents invalid data entry.

Types of SQL Constraints:

1. NOT NULL:

Ensures that a column cannot have a NULL value.

2. UNIQUE:

Ensures that all values in a column are distinct.

3. PRIMARY KEY:

Combines NOT NULL and UNIQUE. Identifies each record uniquely in a table.

4. FOREIGN KEY:

Establishes a relationship between two tables by linking a column to the primary key of another table.

5. CHECK:

Ensures that all values in a column satisfy a specific condition.

6. DEFAULT:

Sets a default value for a column if no value is provided.

7. INDEX:

Improves query performance by creating an index on a column.